Benchmark Analysis Report

Analytics

Executive Summary

This report presents a comprehensive analysis of benchmark performance across multiple batches and questions.

The analysis includes both success rate metrics and detailed EQUATOR reporting guideline compliance assessment.

Key Performance Metrics:

- Total runs analyzed: 13
- Overall success rate: 76.9%
- Average runtime: 711.5 seconds
- Number of batches: 1
- Number of unique questions: 13

EQUATOR Analysis (All Criteria):

- Total EQUATOR criteria evaluated: 34
- Average compliance score: 0.853 (85.3%)

EQUATOR Analysis (Relevant Criteria):

- Relevant EQUATOR criteria evaluated: 30
- Average compliance score: 0.917 (91.7%)

Note: The relevant criteria analysis excludes items not typically included in analytics reports, such as funding information and certain methodological details not applicable to our use case.

Success Rate and Consistency Analysis

Summary Statistics

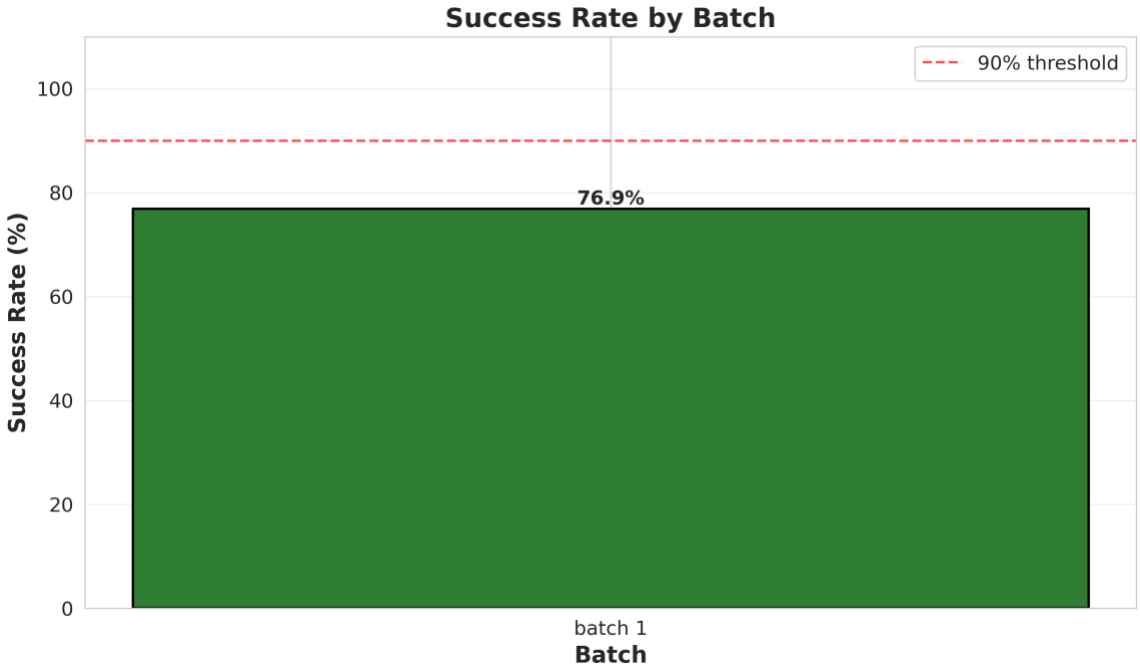
Comprehensive summary statistics for the benchmark analysis, including key performance metrics and distributions.

Benchmark Summary Statistics

Metric	Value
Total Runs	13
Total Successes	10
Overall Success Rate (%)	76.92
Average Success Rate per Batch (%)	76.92
Std Dev of Batch Success Rates (%)	nan
Average Total Runtime (s)	711.48
Median Total Runtime (s)	569.79
Std Dev Total Runtime (s)	445.22
Min Total Runtime (s)	109.35
Max Total Runtime (s)	1754.46
Average Step 1 Time (s)	4.42
Average Step 2 Time (s)	35.69
Average Step 3 Time (s)	10.02
Average Step 4 Time (s)	7.74

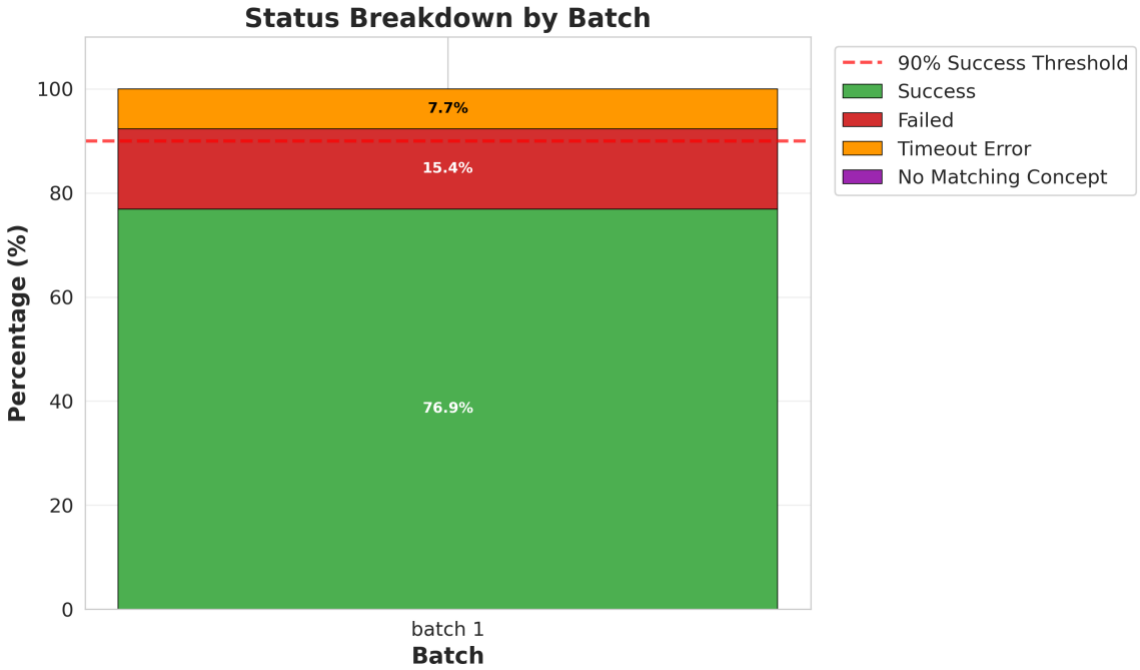
Batch Success Rates

Success rate for each batch with a 90% threshold reference line to identify high-performing batches.



Status Breakdown by Batch

Detailed breakdown of all status types (success, failure, error, timeout) for each batch to identify patterns.



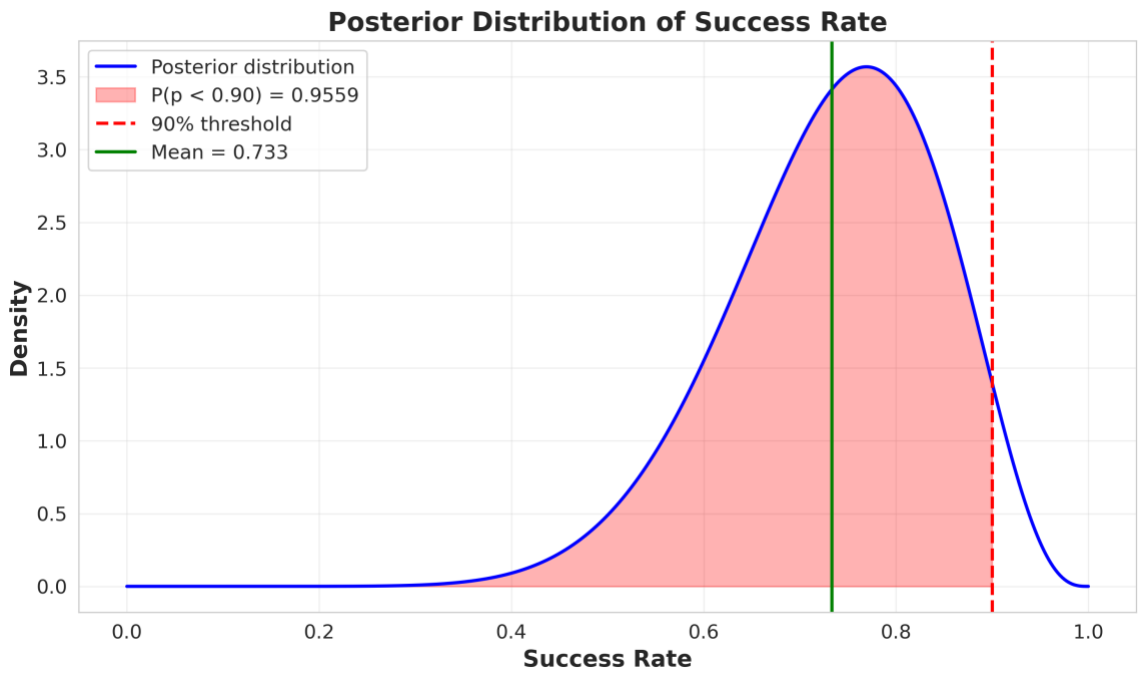
Status Heatmap

Detailed heatmap view of success/failure patterns across all batches and questions, highlighting problematic areas.

Status Heatmap Across Batches and Questions

Bayesian Analysis

Probabilistic assessment of success rate performance using Bayesian inference to estimate confidence intervals.

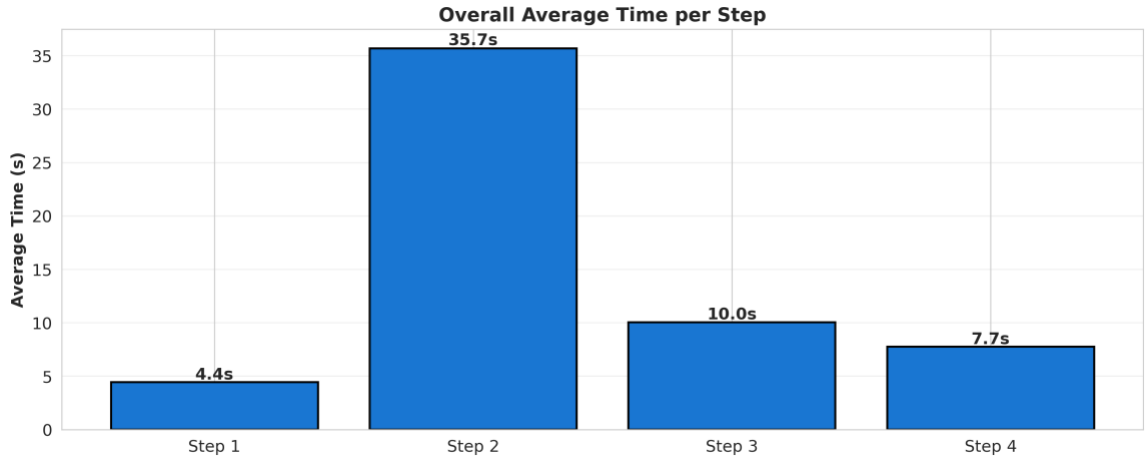


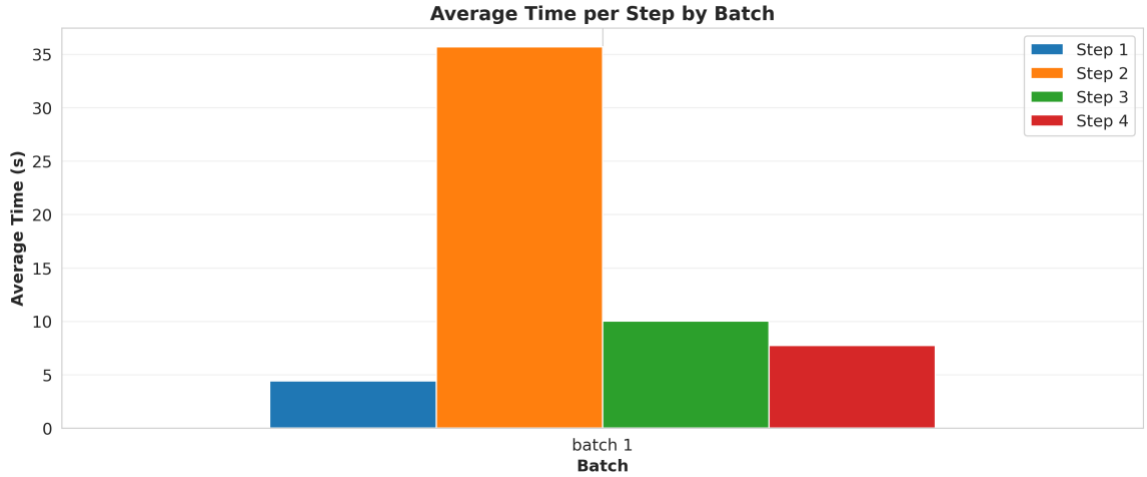
Bayesian Analysis Interpretation

Bayesian Analysis Results:
- Observed successes (x): 10
- Total runs (n): 13
- Beta distribution: Beta(11, 4)
- P(success rate < 90%): 0.9559
- Posterior mean: 0.733

Batch Runtime Analysis

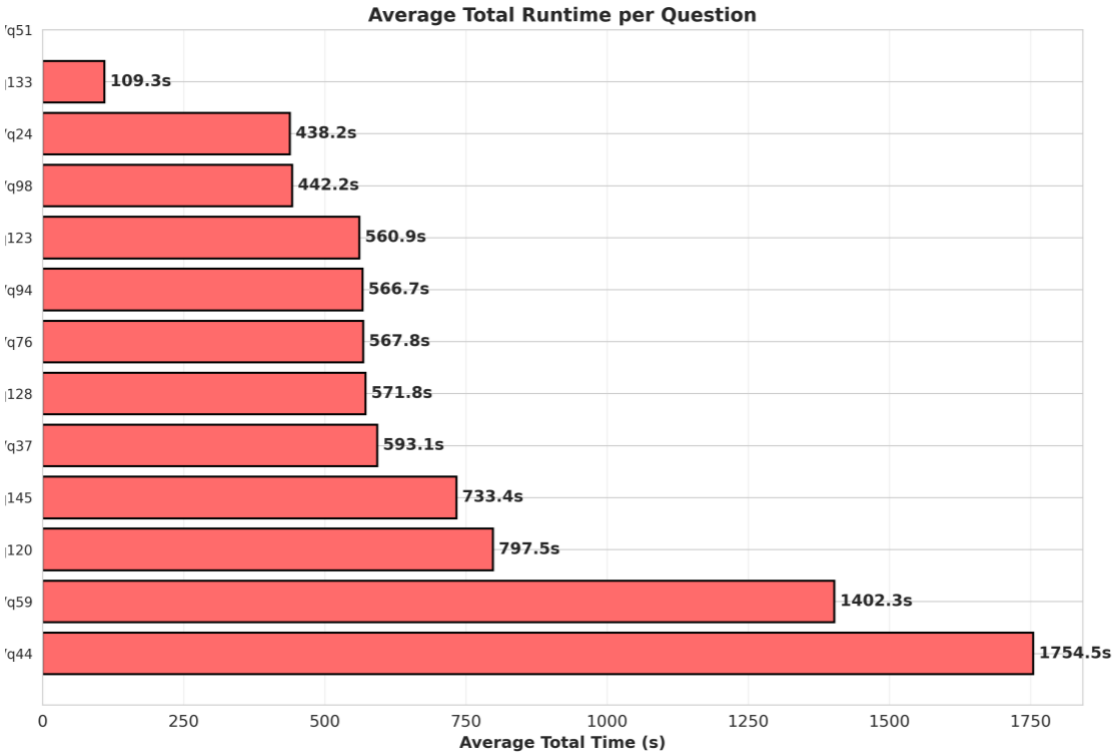
Detailed analysis of runtime performance across all batches, showing distribution and outliers.





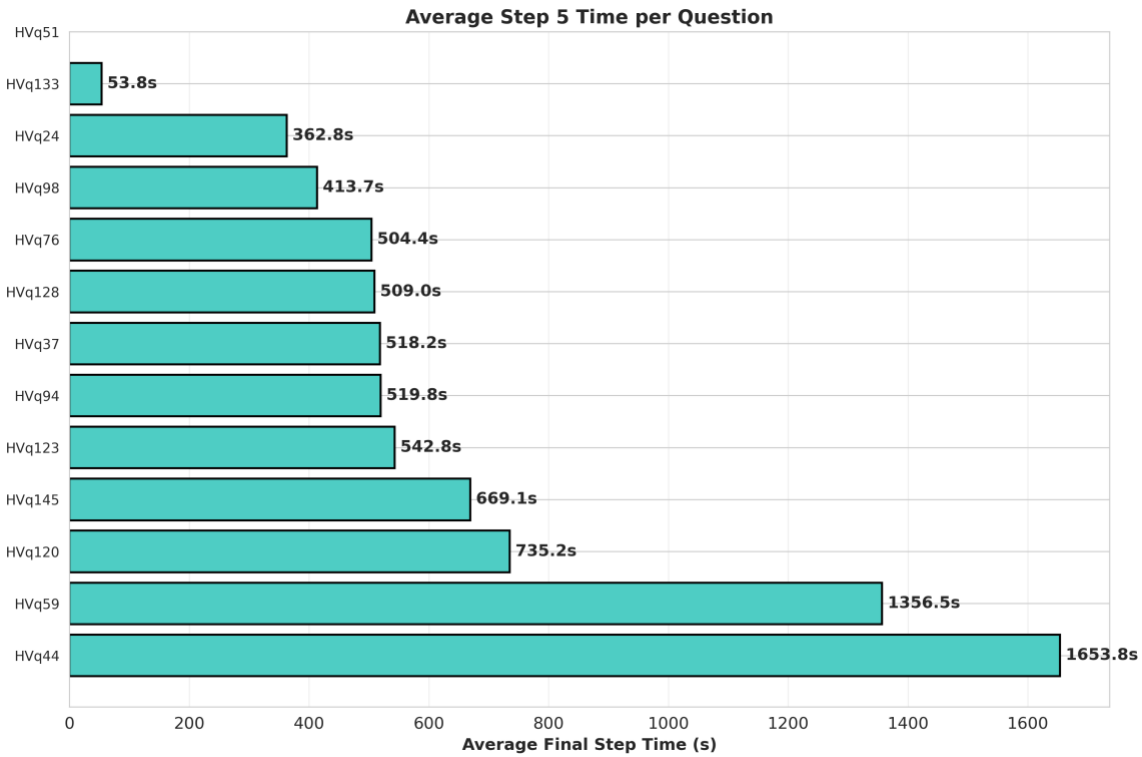
Total Question Runtime Analysis

Runtime performance analysis for each question across all batches, identifying computationally intensive questions.



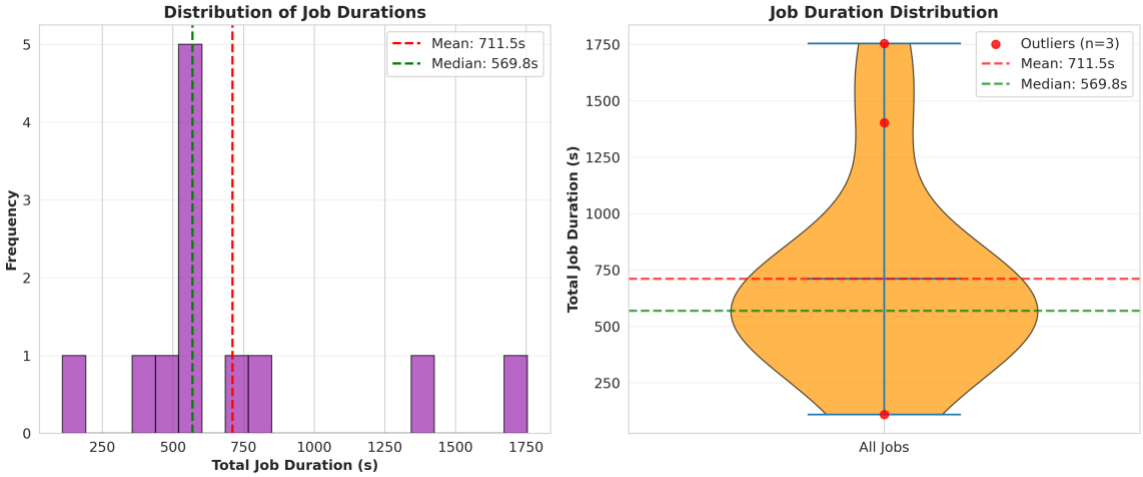
Final Step Runtime Analysis

Analysis of final step (step 5) runtime performance, focusing on the most critical execution phase.



Runtime Distribution Analysis

Statistical distribution of runtime values with outlier detection and performance benchmarks.



Alignment with EQUATOR Criteria (All)

This section presents results comparing our outputs to the entire EQUATOR checklist with no exclusions.

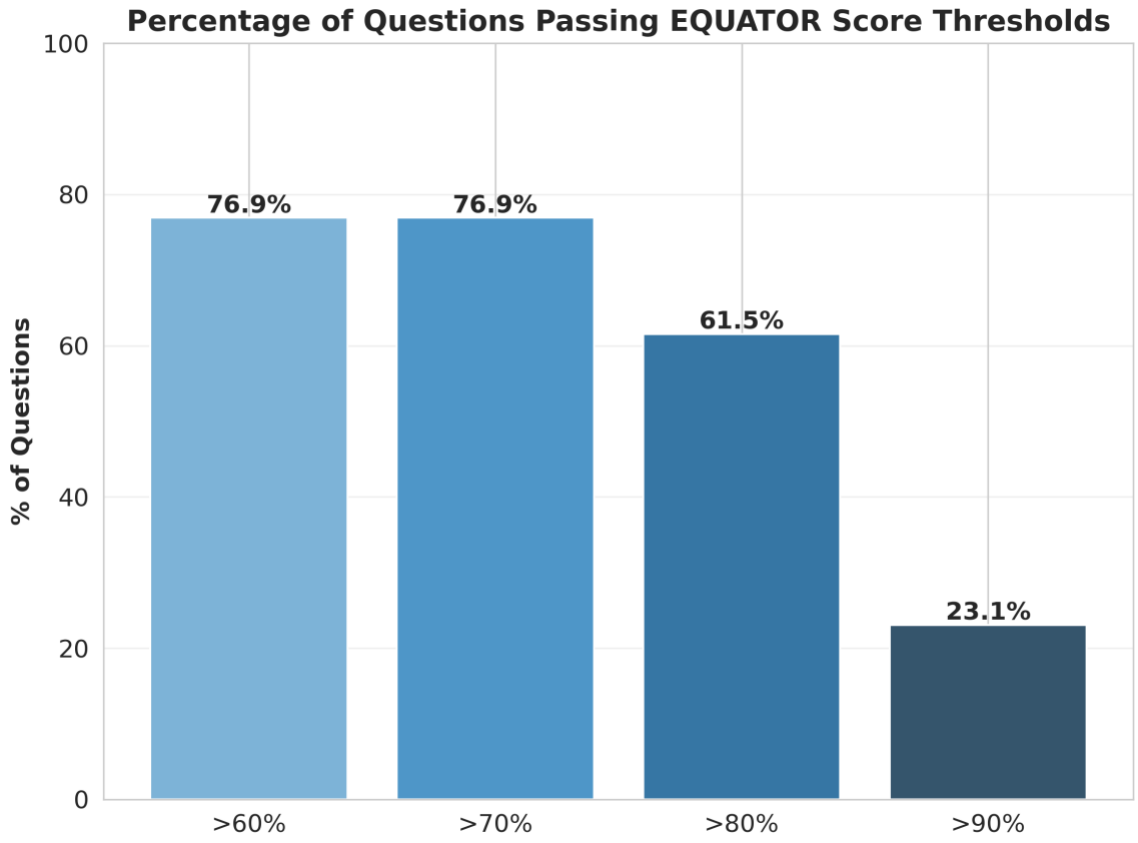
Scores may be lower than expected because certain criteria (such as funding information, specific

methodological details, and administrative elements) are not provided in standard analytics

reports by design.

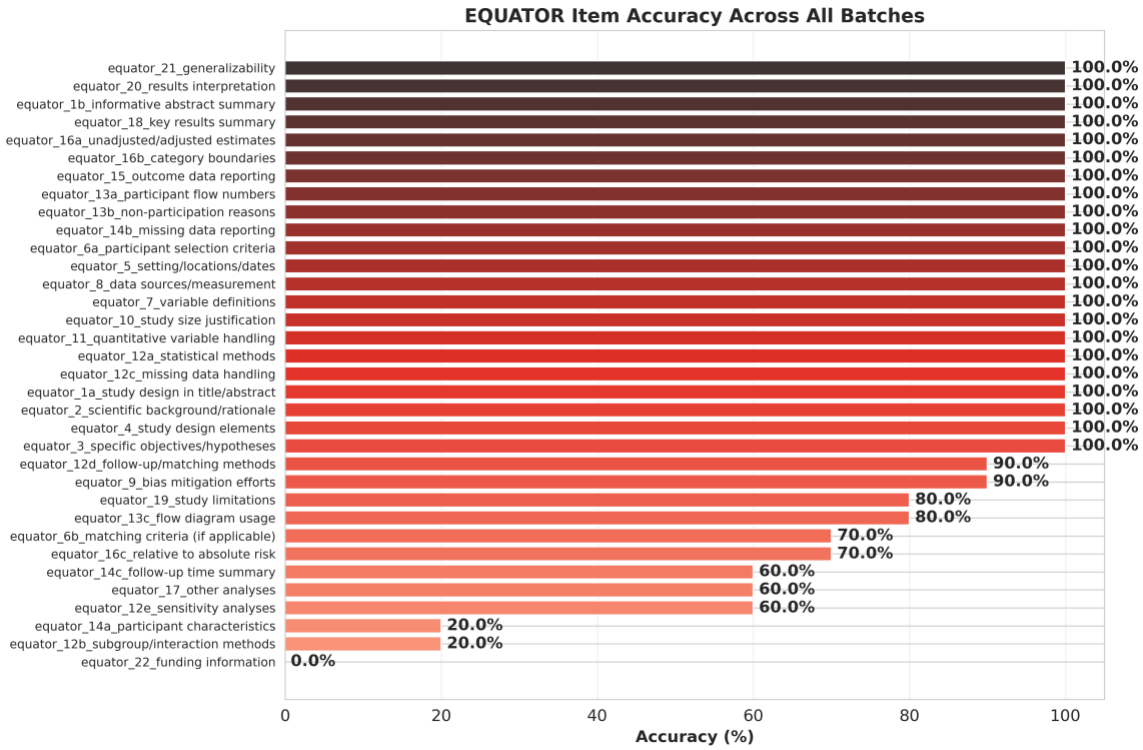
EQUATOR Score Thresholds

Percentage of questions meeting various EQUATOR score thresholds, showing overall compliance distribution.



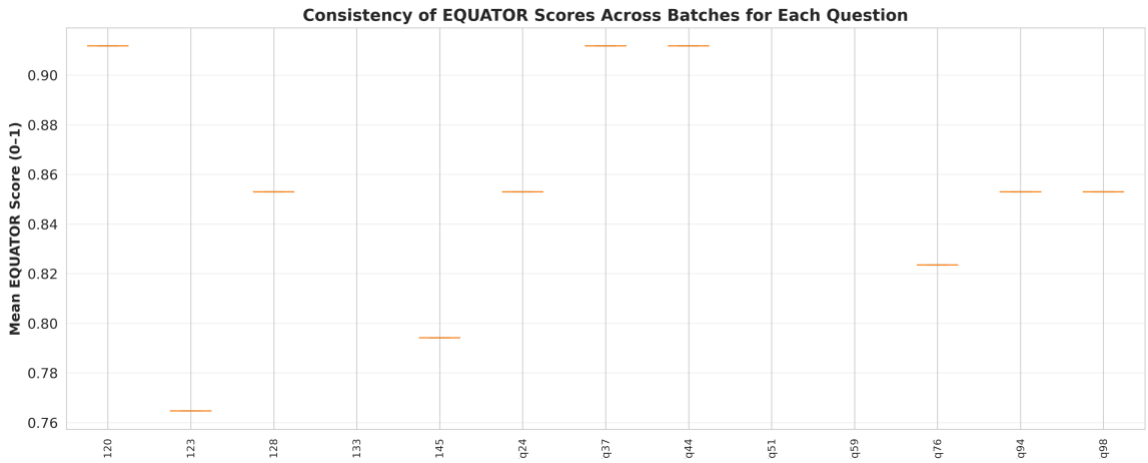
EQUATOR Criteria Accuracy

Average accuracy for each individual EQUATOR criterion, identifying strengths and weaknesses in reporting.



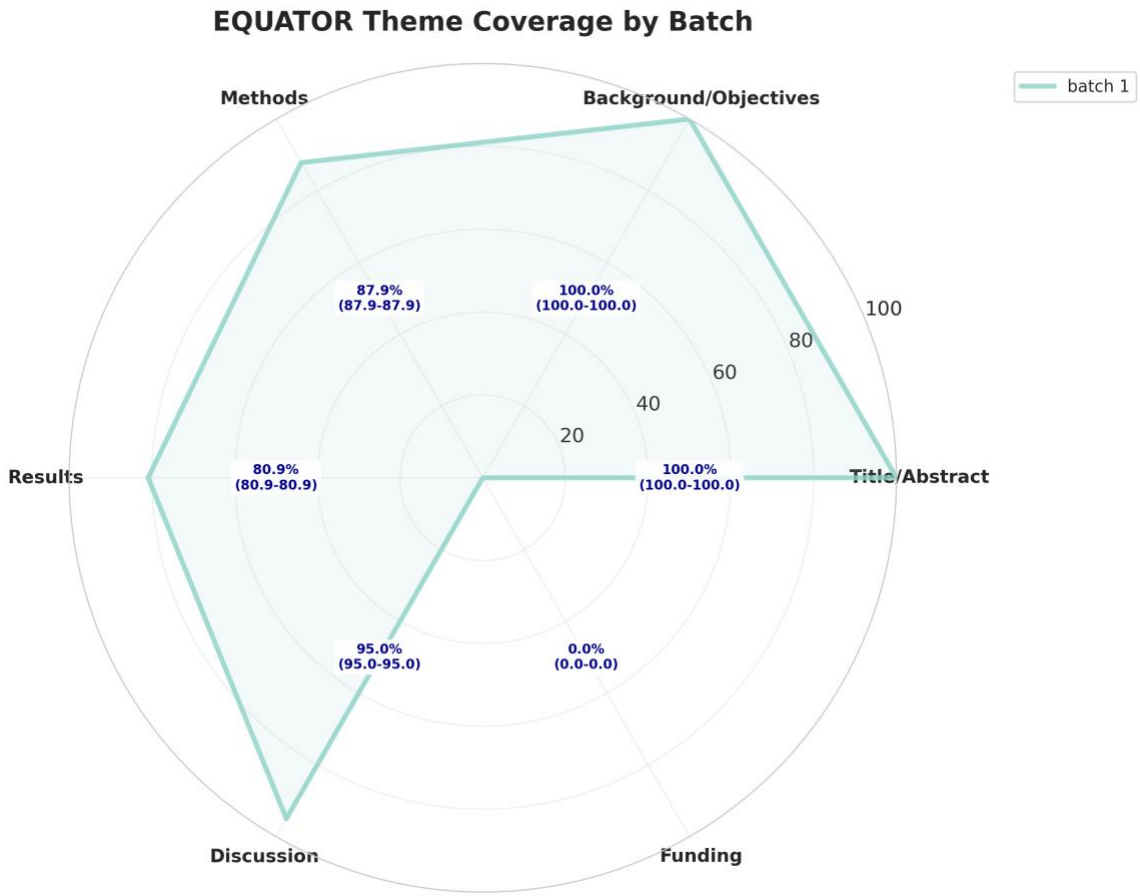
EQUATOR Score Consistency

Variability of EQUATOR scores for each question across different batches, measuring consistency.



EQUATOR Theme Coverage by Batch

Radar chart comparing EQUATOR theme coverage across all batches, highlighting reporting area performance.



EQUATOR Analysis Summary

The EQUATOR (Enhancing the Quality and Transparency of Health Research) analysis evaluates
compliance with reporting guidelines across 34 different complete EQUATOR checklist.

Key EQUATOR Findings:

- Average compliance score: 0.853 (85.3%)
- Questions with high compliance (≥80%): 8/13 (61.5%)
- Criteria range from study design elements to methodological reporting standards
- Analysis reveals variation across different reporting themes (Methods, Results, Discussion, etc.)

The radar chart analysis reveals batch-specific strengths and weaknesses across

different

EQUATOR themes, helping identify targeted areas for improvement in research reporting quality.

Alignment with EQUATOR Criteria (Relevant)

This section focuses on EQUATOR criteria that are directly relevant to analytics reports.

The following criteria have been excluded as they are not applicable to our standard reporting format:

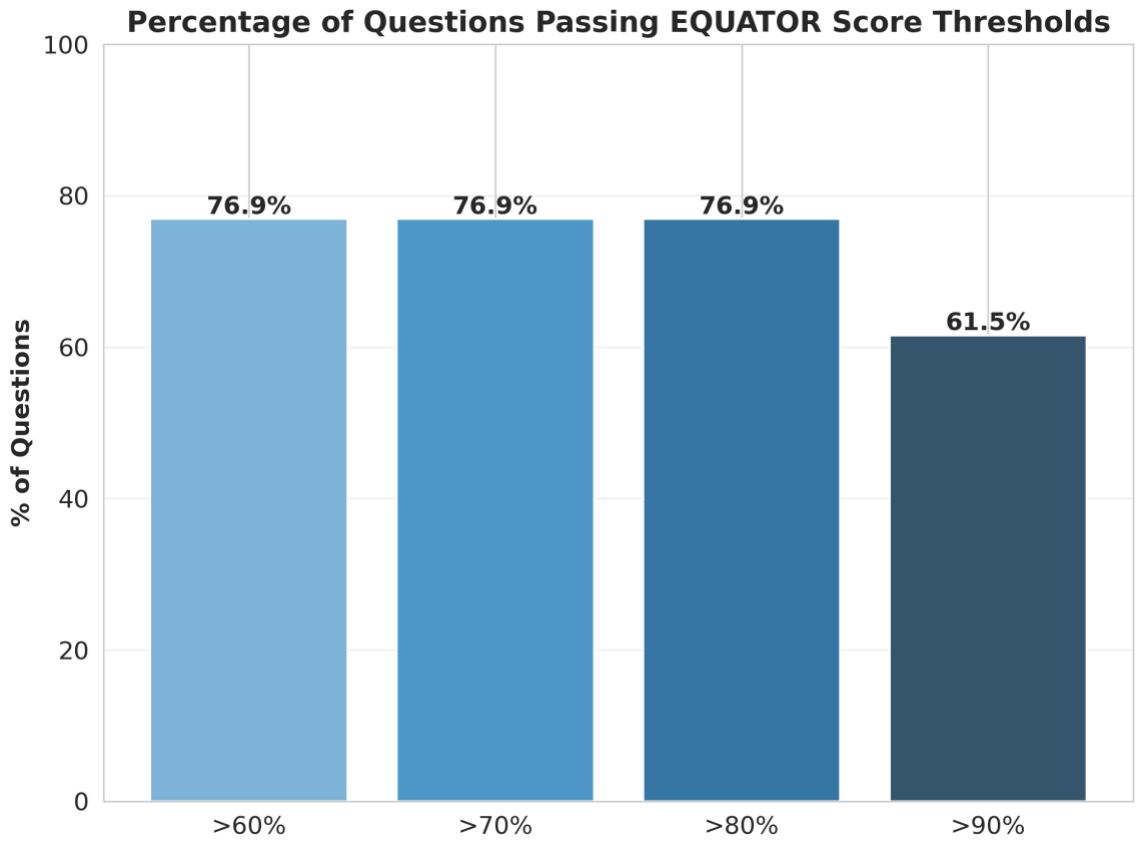
Excluded Criteria:

- equator_6b_matching criteria (if applicable)
- equator_12b_subgroup/interaction methods
- equator_12e_sensitivity analyses
- equator_22_funding information

This focused analysis provides a more accurate assessment of our reporting quality for applicable criteria.

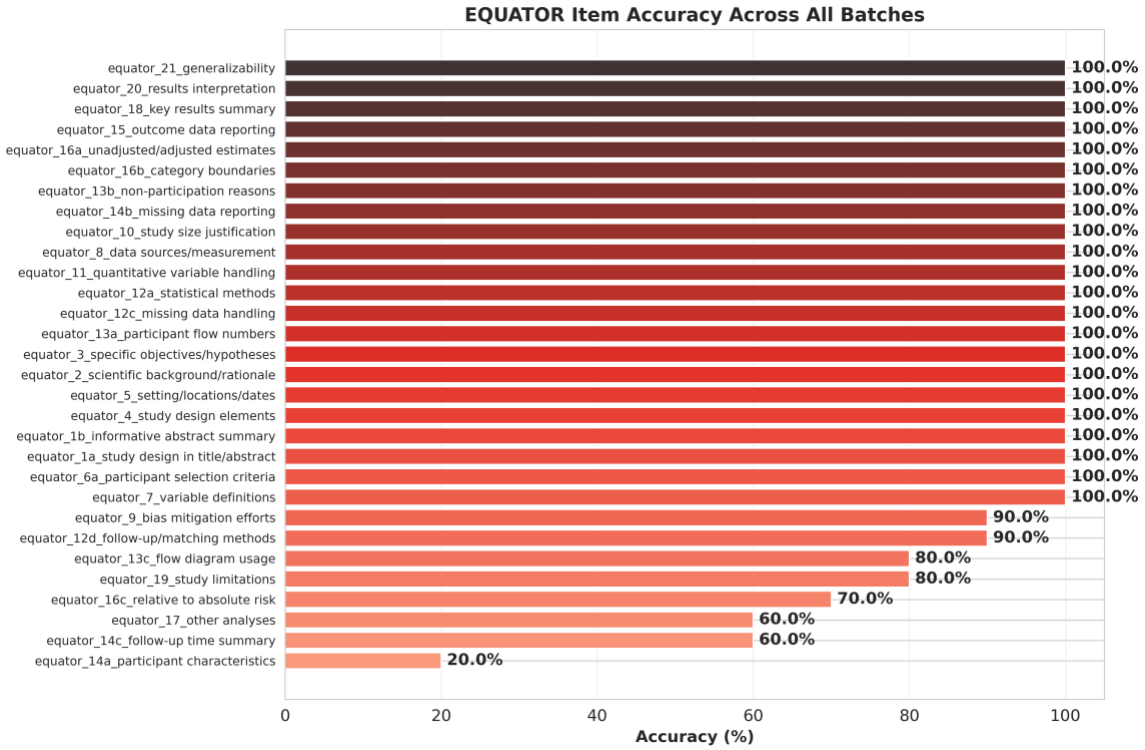
Relevant EQUATOR Score Thresholds

Percentage of questions meeting various EQUATOR score thresholds for relevant criteria only.



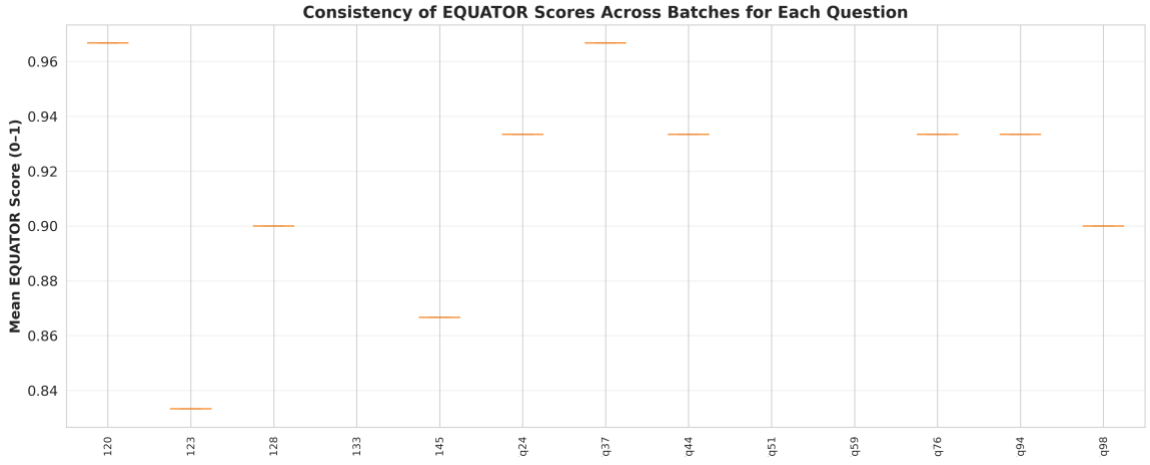
Relevant EQUATOR Criteria Accuracy

Average accuracy for each relevant EQUATOR criterion, showing targeted performance metrics.



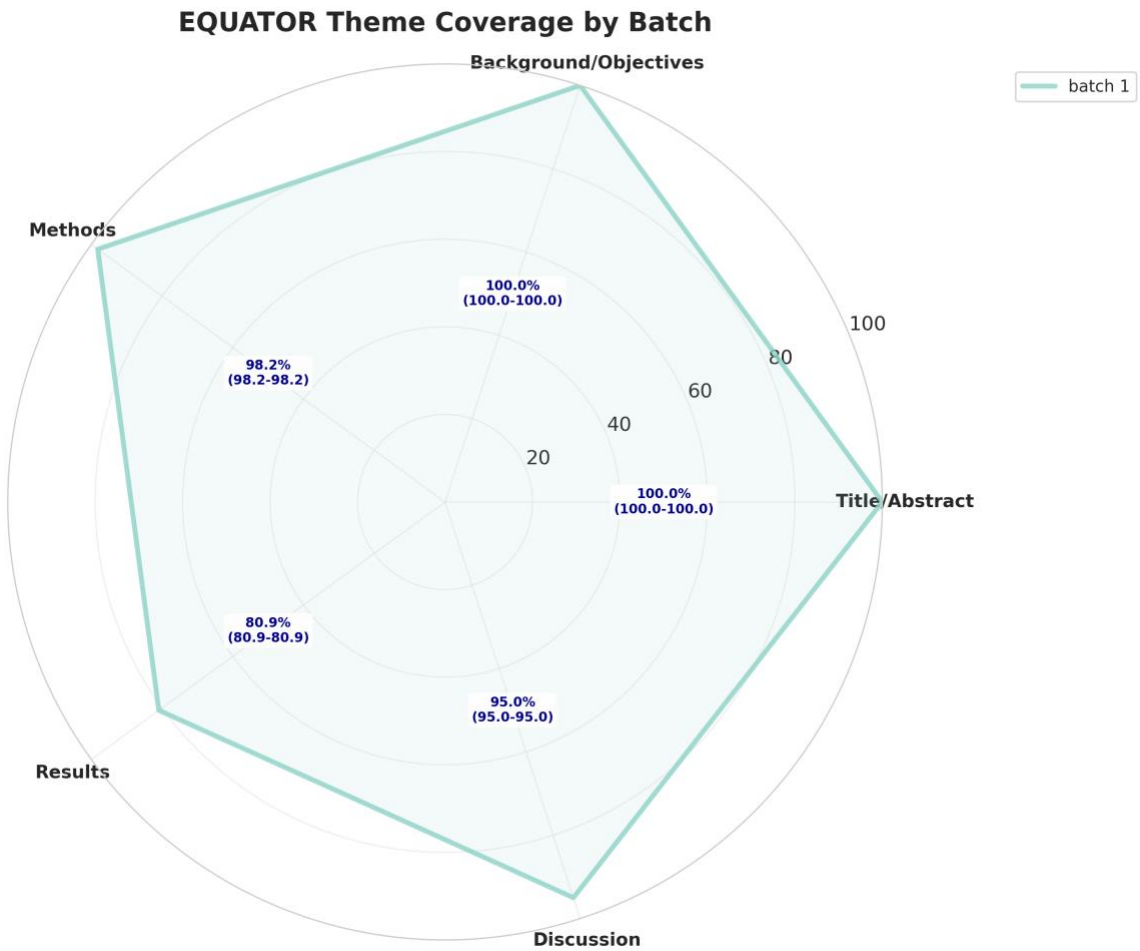
Relevant EQUATOR Score Consistency

Variability of relevant EQUATOR scores across batches, measuring consistency for applicable criteria.



Relevant EQUATOR Theme Coverage by Batch

Radar chart comparing relevant EQUATOR theme coverage, focusing on applicable reporting areas.



EQUATOR Analysis Summary

The EQUATOR (Enhancing the Quality and Transparency of Health Research) analysis evaluates
compliance with reporting guidelines across 30 different relevant EQUATOR criteria.

Key EQUATOR Findings:
• Average compliance score: 0.917 (91.7%)
• Questions with high compliance (≥80%): 10/13 (76.9%)
• Criteria range from study design elements to methodological reporting standards
• Analysis reveals variation across different reporting themes (Methods, Results, Discussion, etc.)

The radar chart analysis reveals batch-specific strengths and weaknesses across different
EQUATOR themes, helping identify targeted areas for improvement in research reporting quality.